

BIOS 735: Bayesian Linear Regression

Alexander McLain

A Bayesian View of Regression

- Suppose we model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I).$$

- In a frequentist analysis, β is an unknown but fixed parameter.
- In a Bayesian analysis, we treat β as a random quantity and specify a **prior distribution**:

$$\beta \sim p(\beta).$$

- We then combine the prior with the likelihood:

$$\underbrace{p(\beta \mid Y)}_{\text{Posterior}} \propto \underbrace{p(Y \mid \beta)}_{\text{Likelihood}} \underbrace{p(\beta)}_{\text{Prior}}.$$

- Bayesian inference combines what we observe in the data with what we believe about the parameters *before seeing the data*

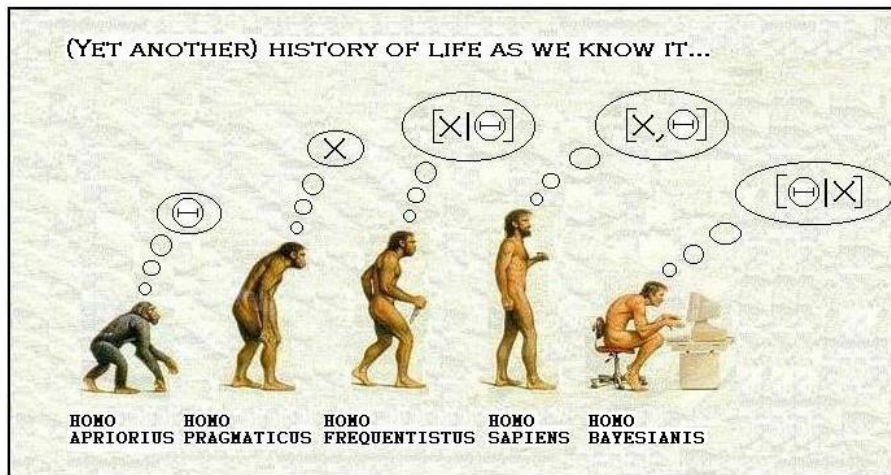


Figure: “A Bayesian is one who – vaguely expecting a horse – and catching a glimpse of a donkey, strongly believes he has seen a mule.”

Ridge Regression

Ridge regression estimates β by minimizing

$$\sum_{i=1}^n (y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2.$$

The second term is the **penalty**.

$$\lambda \sum_{j=1}^p \beta_j^2$$

It shrinks the regression coefficients toward zero.

Ridge Regression as a Bayesian Model

Now suppose we place independent normal priors on the coefficients:

$$\beta_j \stackrel{\text{ind}}{\sim} N(0, \tau^2).$$

The posterior mode is obtained by minimizing

$$\sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2 + \frac{\sigma^2}{\tau^2} \sum_{j=1}^p \beta_j^2.$$

Therefore,

$$\lambda = \frac{\sigma^2}{\tau^2}$$

Connection

The ridge estimator is the **Bayesian posterior mode** under normal priors centered at zero.

Penalized Regression and Bayesian Priors

The ridge connection is not unique.

Penalty	$\longleftrightarrow$	Prior distribution
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Method	Penalty	Prior
Ridge	$\sum_j \beta_j^2$	Normal
Lasso	$\sum_j \beta_j $	Laplace
Elastic net	$\lambda_1 \sum_j \beta_j + \lambda_2 \sum_j \beta_j^2$	Combination

What Makes Bayesian Methods Powerful?

- ▶ A prior does more than shrink coefficients.
- ▶ It gives us a way to incorporate **information beyond the current dataset**.
- ▶ For example, previous studies may tell us that a biomarker is probably associated with disease risk, and that the association is likely to be positive.
- ▶ We can encode that information through the prior distribution for β .

Example: An Informative Prior

Suppose previous biological studies suggest:

- ▶ $\beta > 0$ and suggest that an effect around 0.5 is plausible
- ▶ many of the variables have zero effect (sparsity assumption)

We might specify a **spike-and-slab** prior:

$$\beta_j \sim s_j \delta_0 + (1 - s_j) N(0.5, 0.25^2)$$

where $s_j \in \{0, 1\}$ with $P(s_j = 1) = \pi$.

This prior says:

- ▶ approximately π coefficients are non-zero.
- ▶ positive effects are more plausible than negative effects;
- ▶ effects near 0.5 are most plausible;

We've Already Been Encoding Information

- ▶ This isn't a new idea for us — we've already been doing a version of it all semester.
- ▶ Penalized regression already builds outside knowledge into a model — through the penalty, not the data.
 - ▶ **Grouped lasso:** “these predictors belong together — if one matters, the group matters.”
 - ▶ **Fused lasso:** “neighboring coefficients should be similar — effects change gradually, not randomly.”
- ▶ In both cases, the penalty is a stand-in for something we believe about the world, applied *before* we look at the outcome.
- ▶ A Bayesian prior does exactly this job — but it isn't limited to “shrink” or “fuse.” It's a general language for saying what we already believe, and how strongly.

Beyond What a Penalty Can Say

A lasso-type penalty can say “small,” “grouped,” or “smooth.” A prior can say almost anything you actually believe:

- ▶ **Direction:** this effect should be positive, not just small.
- ▶ **External evidence:** a prior study estimated this effect at roughly 0.5 — start from there instead of zero.
- ▶ **Confidence:** how strongly do we trust that outside evidence? A tighter or wider prior encodes that directly.
- ▶ **Structure across groups:** coefficients can be encouraged to be similar *and* allowed to differ, possibly across space and time.

The grouped and fused lasso are two specific cases of using prior information. A prior is the more general tool: any belief you can state in words, you can usually build into a prior.

Bayesian Application: Borrowing from Historical Trials

- ▶ Suppose we are conducting a new randomized trial for Acute myeloid leukemia (AML) with a relatively small concurrent control group.
- ▶ Previous AML trials contain **relevant control-arm data**.

Historical data $\longrightarrow$ Informative prior

- ▶ The prior provides information about the expected outcome in the control group, while the current trial provides the new evidence.
- ▶ Rather than treating the historical data as either completely irrelevant or completely equivalent to the current data, Bayesian methods can **borrow information while accounting for differences between trials**.

van Rosmalen, Joost, et al. "Including historical data in the analysis of clinical trials: Is it worth the effort?." *Statistical methods in medical research* 27.10 (2018): 3167-3182.

Prior Information + Data

- ▶ The prior does **not** determine the final estimate.
- ▶ The posterior combines the prior with the likelihood:

$$p(\beta | Y) \propto p(Y | \beta) p(\beta)$$

Prior information + **Observed data inference** → **Posterior**

- ▶ Bayesian methods give us a formal framework for incorporating **external scientific knowledge** into a statistical model.
- ▶ How much that external knowledge plays a role depends on the strength of the prior (similar to the amount of penalization).

Bayesian Linear Regression: Advantages

1. **Probabilistic Interpretation:** Bayesian methods provide a full posterior distribution over the model parameters. This allows for natural quantification of uncertainty about variable importance, coefficient estimates, and predictions.
2. **Flexible Priors:** Bayesian frameworks allow for the inclusion of various prior distributions, which can be informed by domain knowledge.
3. **Natural Incorporation of Hierarchical Structures:** Hierarchical Bayesian models can be easily formulated to share information across groups of related variables, aiding in variable selection when there's a known structure in predictors.
4. **Variable Selection with probabilities.**
5. **Model Averaging:** Bayesian methods allow for model averaging over a range of models, weighing them by their posterior probabilities.

Disadvantages and R packages

- ▶ The major disadvantages to Bayesian methods:
 - ▶ Can be sensitive to the prior distribution
 - ▶ Computationally intensive.
- ▶ Bayesian linear regression techniques can be fitted in R (among others):
 - ▶ **VB packages:** sparsevb, and varbvs,
 - ▶ **MAP estimation via EM:** probe, EMVS, SSLASSO, and lmmprobe,
 - ▶ **Full MCMC:** ebreg, horseshoe,



**Yesterday's
posterior is
today's prior**